



CSE260: Digital Logic Design

Spring 2025

Quiz - 02

Duration: 30 Minutes

A

Name: *Solution*

ID:

Section:

Instructions: Answer on the space provided.

Question 1 [CO2]: Convert the boolean function to its Canonical SOP form. [7.5]

$$F(A,B,C) = C + B'C + AC' + BC'$$

$$= C(1+B') + AC'(B+B') + BC'(A+A')$$

$$= C(A+A')(B+B') + ABC' + AB'C' + ABC' + A'BC'$$

$$= (AC + A'C)(B+B') + ABC' + AB'C' + ABC' + A'BC'$$

$$= ABC + AB'C + A'BC + A'B'C + \underline{ABC'} + \underline{AB'C'} + \underline{ABC'} + \underline{A'BC'}$$

$$= ABC + AB'C + A'BC + A'B'C + ABC' + AB'C' + A'BC'$$

$$= 111 + 101 + 011 + 001 + 110 + 100 + 010$$

$$= \Sigma(7, 5, 3, 1, 6, 4, 2)$$

$$= \Sigma(1, 2, 3, 4, 5, 6, 7)$$

Question 2 [CO1]: Draw the following functions using NOR gates only.

[7.5]

$$F(A,B,C) = A' + B \odot C$$

$$= A' + BC + B'C'$$

